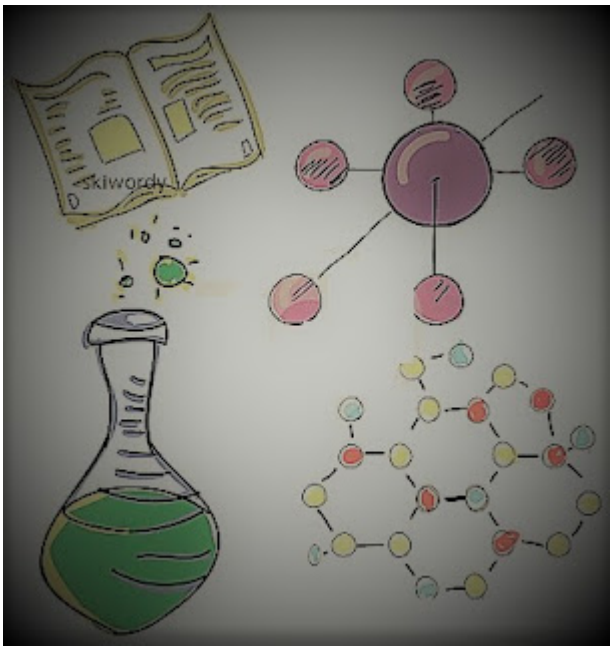
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CHEMISTRY CHAPTER 1 "STOICHIOMETRY" QUESTIONS AND ANSWERS CLASS 11



CHEMISTRY NATIONAL BOOK FOUNDATION

CHAPTER 1

"STOICHIOMETRY"

EXERCISE QUESTIONS AND ANSWERS:

Q: 58.5 amu are termed as formula mass and not molecular mass of NaCl. Why?

Ans: As NaCl is an ***ionic compound***. Ionic compounds often form ***crystals***. Also, they cannot exist as ***individual, neutral, and independent molecules.*** Hence, term formula mass is used to express the mass of ionic compound instead of molecular mass. **This is the reason 58.5 amu are termed as formula mass and not molecular mass of NaCl.**

Q: Concept of limiting reactant is not applicable to the reversible reactions. Explain.

Ans: As in reversible reactions, reactants are not completely converted into products. There exist a dynamic equilibrium, reactants are converting into products and products into the reactants at the constant reaction rate. ***The reactant which is consumed earlier and completely in a chemical reaction is called limiting reactant.*** As reactants, in reversible reactions, is ***not consumed completely***. Therefore, the concept of limiting reactant is not applicable to the Reversible reactions.

Q: How many covalent Bonds are Present in 9g of H2O?

Ans:

Mass of Water	9g
Molar Mass of Water=	18 g/mol
Formula No. of molecules=	Mass of H2O/ Molar Mass of H2O x 6.02 x 1023
No. of molecules=	9/18 x 6.02 x 1023 3.01 x 1023 molecules
1 molecule of H2O contains covalent bonds =	2 bonds
3.01 x 1023 molecules of H2O contains covalent Bonds=	2 x 3.01 x 1023 Bonds 6/02 x 1023 bonds

Q: Differentiate Between Limiting and Non-Limiting Reactants.

Ans:

Limiting Reactant	Non-Limiting Reactant
It is consumed completely in a chemical reaction.	It is left unused and un-reacted after completion of reaction.
It controls the amount of product formed.	It does not controls the amount of product formed.
It is consume earlier in a chemical reaction	It does not consume earlier in a chemical reaction
The amount of product Formed depends on the initial amount of limiting reactant.	The amount of product Formed indepedent of the initial amount of non-limiting reactant.

Q: How many molecules of water are there in 12 g of ice?

Ans:

Mass of ice	12 g
Molar Mass of ice=	18 g/mol
Formula No. of molecules=	Mass of H2O/ Molar Mass of H2O x 6.02 x 1023
No. of molecules=	12/18 x 6.02 x 1023
	4.01 x 1023 molecules

Q: One Mole of H2SO4 should completely react with 2 moles of NaOH . How does Avagadro's number help to explain it?

Ans:

Avogadro's number is a constant that represents the number of particles (atoms, molecules, or ions) in one mole of a substance. The value of Avogadro's number is approximately 6.02 x 10^23 particles per mole.

In the case of the reaction between one mole of H2SO4 and 2 moles of NaOH, Avogadro's number helps to explain why the balanced equation for the reaction is:

The chemical Reaction of H2SO4 and NaOH is

2NaOH + H2SO4 ----------> Na2SO4 + 2H2O

2 Formula Units + 1 Molecule ------> 1 Formula Unit + 2 Molecules

2 Moles + 1 Mole ------> 1 Mole + 2 Mole

Thus, according to Avogadro's the above equation shows that 2 x 6.02 x 1023 formula units of NaOH should react with 1X 6.02 x 1023 molecules of H2SO4 to form 1X 6.02 x 1023 Formula units of Na2SO4 and 2 x 6.02 x 1023 molecules of water for a **balanced chemical equation.**

In summary, Avogadro's number helps to explain why the balanced equation for the reaction between one mole of H2SO4 and 2 moles of NaOH is as it is, by allowing us to calculate the number of molecules of each substance involved in the reaction.

Q: Give reason that 1 mole of different compounds have different masses but have the same number of Molecules.

Ans: One mole of different compounds have same number of molecules but different masses. It is because individual molecules of different compounds have different masses, therefore, equal number of moles of different compounds will also have different masses although same number of molecules. e.g. 1 mole CO2 = 6.02x1023 molecules of CO2= 44 g of CO2

Q: 23 g of sodium and 238 g of uranium have equal number of atoms in them.

Ans: 1 Mole of sodium = 23 g of Sodium 1 Mole of Uranium = 238 g of Uranium The One Mole of different elements contain same Number of atoms in them. So, 23 g of sodium and 238 g of uranium have equal number of 6.02 x 1023 atoms in them.

Q: What is Percentage Composition?

Ans: It can be defined as "The relative amount each element Present in a compound."

Q: Calculate the Percentage Composition Of Glucose?

Ans:

Molar Mass of Glucose =	(12x6)+(12x1)+ (16x6)= 180 g/mol
Formula: **% of an element =**	Mass of element in compound/Molar mass of compound X 100
% of an Carbon	=(72/ 180) x 100=40%
% of an Hydrogen	=(12/ 180) x 100=6.67%
% of an Carbon	=(96/ 180) x 100=53.33%

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Q: The liquid CHBr3 has a density of 2.89 g/dm3. What Volume
should be measured to contain a total of 4.8 x 1024 molecules
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Ans: Mass of CHbr3 = ? Molar Mass of CHBr3 = 252.7g/mol No. of molecules of CHBr3 = 4.8 x 1024 Formula No. of molecules= Mass/Molar Mass x 6.02 x 1023 Now: 4.8 x 1024 = Mass/252.7 x 6.02 x 1023 By rearranging Mass= 4.8 x 1024/ 6.02 x 1023 x 252.7 Mass= 2014.88 g Given Density Of CHBr3 = 2.89 g / dm3 Formula d=m/V Putting values V=m/d = 2014.88/2.89 =697.19 dm3

Q: Differentiate between actual and theoretical yield?

Ans:

Actual Yield	Theoretical Yield
The amount of product that is actually produced during a chemical reaction is called actual yield.	It is left unused and un-reacted after completion of reaction.
It controls the amount of product formed.	It does not controls the amount of product formed.
It is consume earlier in a chemical reaction	It does not consume earlier in a chemical reaction
The amount of product Formed depends on the initial amount of limiting reactant.	The amount of product Formed indepedent of the initial amount of non-limiting reactant.

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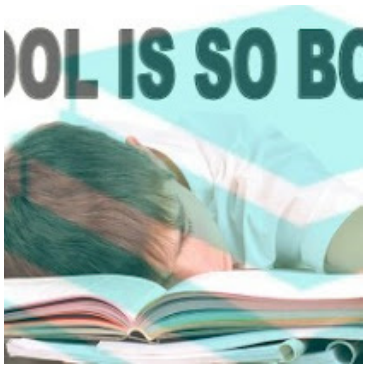
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6 comments:

Unknown — September 2, 2018 at 3:48 PM

Perfect..!!

REPLY

Unknown — August 22, 2019 at 9:51 PM

Thnku so mch

REPLY

Unknown

September 2, 2019 at 12:35 AM

This is very helpful but i can't understand Qone mole ofH2SO4 when react..... 😓

REPLY

Unknown

February 23, 2020 at 10:49 PM

As stated by Stanford Medical, It is really the SINGLE reason this country's women live 10 years more and weigh on average 19 kilos less than we do.

(And by the way, it has NOTHING to do with genetics or some secret exercise and EVERYTHING to about "how" they are eating.)

BTW, I said "HOW", not "what"...

TAP this link to see if this easy quiz can help you release your true weight loss potential

REPLY

Anonymous

August 6, 2021 at 7:13 PM

Thanks very much

REPLY

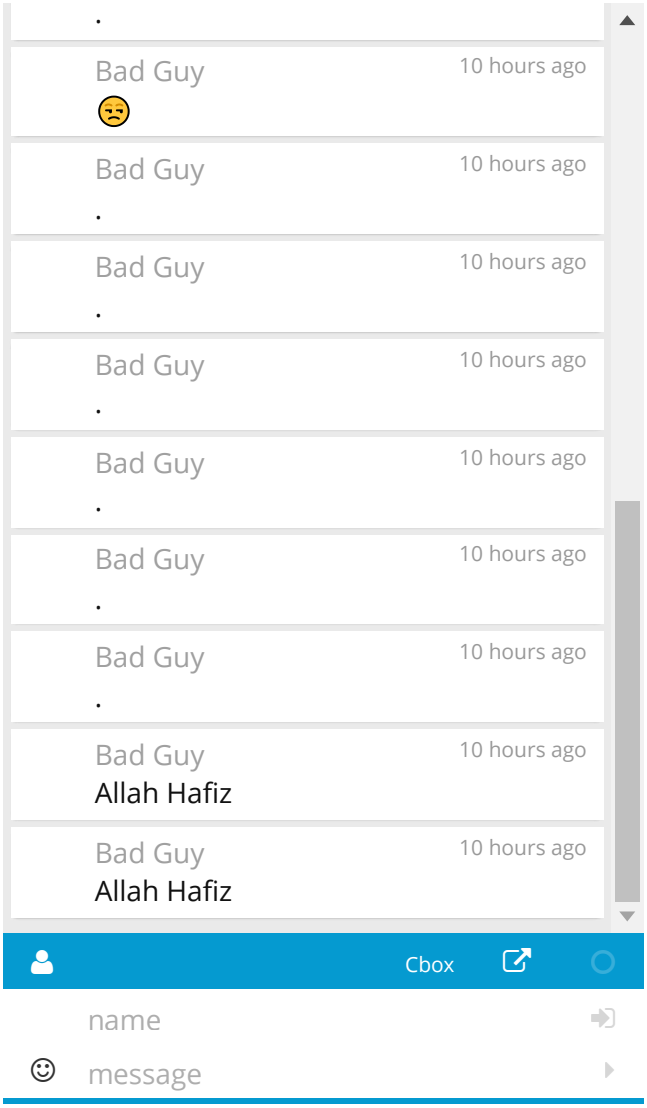
Anonymous

October 1, 2022 at 4:34 PM

Appreciate it

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